

Network Models Problem Set

Problem 1: For the WS model with probability p the key plots are the average clustering

$$\frac{\langle C(p) \rangle}{\langle C(0) \rangle}$$

and average path length

$$\frac{\langle d(p) \rangle}{\langle d(0) \rangle}$$

versus p .

(i) What is the clustering coefficient, $C(0)$, for a WS network where every node is connected to nearest and next nearest neighbours?

(ii) For a WS model where only nearest neighbours are connected, $k = 2$, a cycle, what is the average path length at $p = 0$ in a model with $N \gg 1$ nodes?

$$\text{Hint: } 1 + 2 + 3 + \dots + N = \frac{N(N+1)}{2}$$

(iii) Produce the plot below by averaging over many WS networks for a range of values of p .

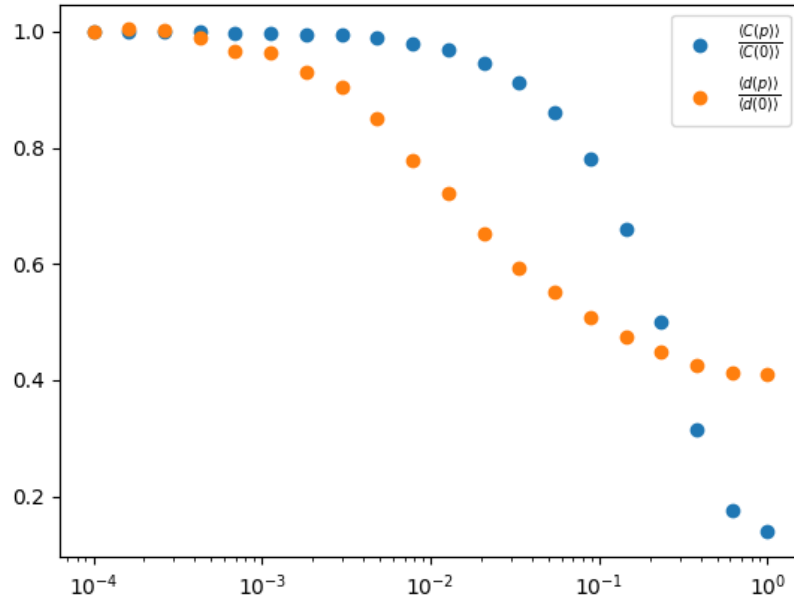


Figure 1: Computed using $N = 100$, $k = 10$ and averaging over 100 different networks at each value of p .

Hint: don't write your own code to generate graphs or measure clustering and path length! Use `networkx`.

Problem 2: The **copying model** consists of the following steps. Starting with some network G (say 2 nodes connected by a single link) add a node and

- Pick a node u at random
- With probability p connect the new node to i
- With probability $(1 - p)$ copy one of u 's connections

Implement this and plot the degree distribution of the network after running the model for a large number of steps.

Problem 3: A configuration model matches k_i stubs on each node. For some given degree sequence k_i show the number of different networks which can be made from that degree sequence (allowing self edges and multi edges) is

$$\frac{2L!}{(2^L)L!}$$

Problem 4: Is the average clustering of the karate club unusual for a network with that degree sequence?

Hint: Compute a distribution by sampling networks from the configuration model.

Problem 5: At $p = 0$ an ER network has every node disconnected, so has N components. At $p = 1$ it has every node connected so has 1 component. This problem looks at approximating the value of p where this transition happens. The **giant component** of a random network is one that contains a large fraction of the nodes fN .

Let q be the probability that a randomly selected node is not in the giant component.

(i) What is the probability a randomly selected node i is not connected to another node j ?

(ii) What is the probability that i is connected to j but j is not in the giant component

Since the probabilities above are exclusive (one or the other happens) we add (i) and (ii) to get the probability of a randomly chosen node not being connected to the giant component via node j .

(iii) q is the total probability of not being connected to the giant component via any of the $N - 1$ other nodes in the network. Use the result of the first two parts and the fact that these are independent events to write down an

equation for q .

(iv) The probability for a node to be in the giant component is $s = 1 - q$. Simplify the equation from (iii) and write it in terms of s . In the limit of large N you should get

$$s = 1 - e^{-\langle k \rangle s}$$

Hint 1: $\langle k \rangle = p(N - 1)$

Hint 2: $\ln(1 - x) \simeq -x$ for small x

(v) This equation can only be solved numerically. Using something like `scipy` solve it for a range of $\langle k \rangle$ values and plot s against $\langle k \rangle$. You should get something like below

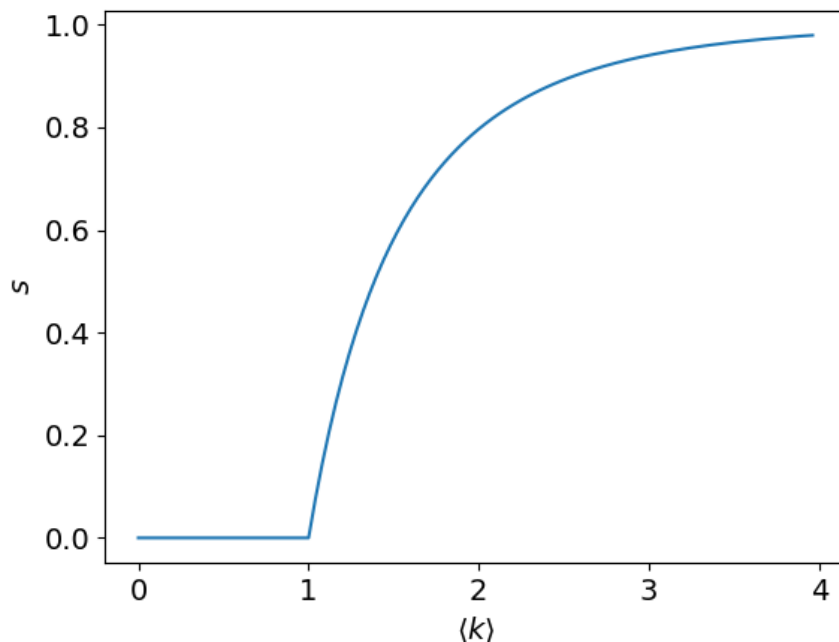


Figure 2: s against $\langle k \rangle$.